

DALY calculation in practice: a stepwise approach

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Received: 23 February 2013 / Revised: 22 March 2014 / Accepted: 3 April 2014 / Published online: 21 April 2014
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Introduction

The disability-adjusted life year (DALY) is an increasingly used population health metric (Murray et al. 2013; Devleesschauwer et al. 2014). DALYs are healthy life years lost, calculated by adding the adjusted number of years lived with disability (YLDs) and the number of years of life lost due to premature mortality (YLLs):

$$\text{YLD} = \text{Number of cases} \times \text{duration till remission or death} \\ \times \text{disability weight} \quad (1)$$

$$\text{YLL} = \text{Number of deaths} \\ \times \text{life expectancy at the age of death} \quad (2)$$

$$\text{DALY} = \text{YLD} + \text{YLL} \quad (3)$$

The philosophical and methodological aspects of the DALY calculation have been described (and debated) in great detail (Murray 1994; Murray and Acharya 1997; for critiques, see, amongst others, Anand and Hanson 1997), and are summarized in the accompanying Hints and Kinks paper (Devleesschauwer et al. 2014). The steps preceding the actual calculation, however, remain less well documented. This Hints and Kinks paper tries to address this gap by presenting a stepwise approach towards a DALY calculation.

Step 1: study population definition

As a first step, the context in which the burden assessment study will take place should be clearly defined. The target population must be delineated by defining study area and

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time period. The latter may be one specific year, or a range of years, which can then be used to calculate the average burden of that time period.

Step 2: disease model definition

The disease model (also called outcome tree) serves as a guide through the further process of the study. Figure 1 presents the causal chain of disease. In general, risk factors increase the risk of disease, either directly or indirectly through facilitating exposure to biological, chemical or physical hazards. The course of disease is characterized by different health states (e.g., acute or chronic phases, short-term or long-term sequelae), possibly having different severity levels. A disease model is a schematic representation of the different health states associated with the concerned cause of disease burden, and the possible transitions between these states.

Depending on the cause of interest, i.e., the disease as such, the hazard or the risk factor, three different approaches may be distinguished:

1. *Outcome-based disease models* represent different health states of diseases, irrespective of the possible (infectious or non-infectious) aetiologies. For example, a disease model for the burden of diarrhoea could describe different diarrhoea severity levels (mild, moderate, severe), contributing YLDs, and diarrhoea-related death, contributing YLLs (Lamberti et al. 2012). Soerjomataram et al. (2012) present generic outcome-based disease models for cancer, including disease progression phases and sequelae.
2. *Hazard-based disease models* represent different health states associated with hazards such as biological or chemical agents or traumas (Mangen et al. 2013).

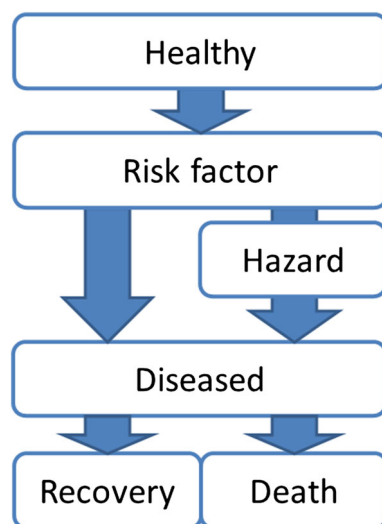


Fig. 1 Generic causal chain of disease

For example, *Campylobacter* infection causes diarrhoea, but also other health states, such as Guillain-Barré syndrome (GBS) (Havelaar et al. 2000). The disease model would thus consist of the different associated symptoms (contributing YLDs), and death attributable to each symptom (contributing YLLs).

3. *Risk factor-based disease models* represent different health states associated with risk factors. For example, a disease model for unsafe water would include the health effects associated with feco-oral pathogens, including *Campylobacter*, other diarrhoeal pathogens and soil-transmitted helminths (Prüss et al. 2002).

A hazard-based disease model for *Campylobacter* infection is given in Fig. 2. This model includes diarrhoea, possibly progressing from acute to chronic stages and to death; GBS occurring in a mild or severe form, the latter being able to cause death; and reactive arthritis. Other disease models can be constructed in a similar way. Disease models can be obtained from previous burden studies. However, new insights might indicate an updated disease model requiring data from a systematic review of relevant clinical and epidemiological aspects.

Step 3: data collection

This step is often the most difficult and time-consuming one. It is also the most crucial one, as the quality of the final DALY estimate directly depends on the quality of the data. Ideally, the necessary data should be collected through a systematic review of peer-reviewed literature and various sources of grey literature, including government agencies, non-governmental organizations and academia. An important guide for conducting and reporting systematic reviews is the PRISMA statement (Moher et al. 2009). As much as possible, collected data should be stratified by age and sex, as this will yield a more precise overall estimate and will enable to study the burden by age and sex. Further stratification by other parameters can also be useful and could allow a breakdown of disease burden by sub-region, occupation, socio-economic status, etc.

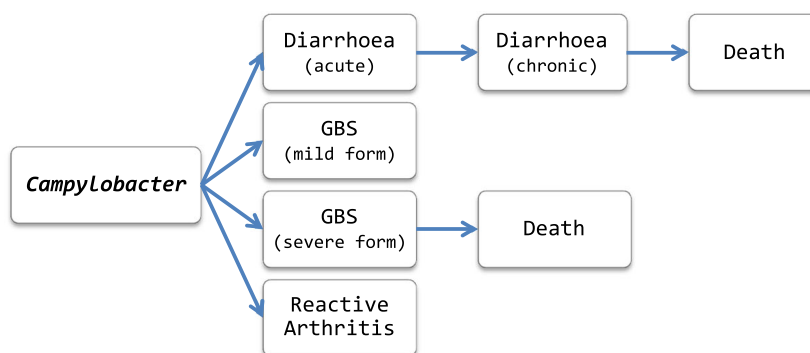
Extrapolation models may be needed when literature searches cannot provide essential data. These models estimate parameters from data of neighbouring regions or other time periods. The external data used must thus be representative of the selected population, region and time. Where no empirical data can be identified, expert elicitation may be applied (see Cooke 1991 for a guide).

In general, three groups of data need to be collected.

Demographical data

DALY calculations require the total number of males and females, per age group, of the selected area and time

Fig. 2 Hazard-based disease model for *Campylobacter* infection (GBS Guillain–Barré syndrome)



period. These data can be obtained from national statistical bureaux or from the United Nations Statistics Division (available from <http://data.un.org/>). A second required source of demographical data is the life-expectancy table. In order to enhance comparability, the life expectancy at the age of death is generally drawn from standard life-expectancy tables (Devleeschauwer et al. 2014). However, local life-expectancy tables may also be used, in particular when within-country disease ranking is an important objective. Local life tables are typically available from national statistical bureaux.

Epidemiological data

The most important data for DALY calculations are the number of cases in the different health states defined by the

state (i.e., regardless of aetiology), and an attribution probability (not applicable for outcome-based disease models):

Overall incidence health state

× proportion attributable to hazard or risk factor

The incidence of GBS due to *Campylobacter* can for instance be obtained by multiplying the overall incidence of GBS in the population to the proportion of *Campylobacter*-attributable GBS. In risk factor-based disease models, the proportion of cases attributable to the risk factor is commonly referred to as the population attributable fraction (Perez and Künzli 2009).

3. *Transition approach* The incidence of a health state can be obtained from the previous incidence in the model. Two scenarios are possible:

Overall incidence burden cause × probability of transition to health state

Incidence health state × probability of transition to next health state

disease model, including death. Most often, the number of incident cases is considered, although DALYs may also be calculated based on the number of prevalent cases (see Schroeder 2012 for a comparison). In the following, we will assume an incidence perspective.

The number of incident cases is the product of the incidence rate and the population size. Depending on data availability, there are three approaches to obtain incidences of the individual health states in the disease model:

1. *Direct approach* The incidence of a health state is directly available, e.g., through a disease or mortality register. In our example (Fig. 1), this would be the case if a prospective population-based study directly estimated the incidence of *Campylobacter*-associated GBS.
2. *Attribution approach* The incidence of a health state is obtained from the overall incidence of the given health

The first scenario would imply that we model the incidence of *Campylobacter*-associated GBS on the overall *Campylobacter* incidence in the population and the probability of developing GBS after *Campylobacter* infection. The second scenario would imply, for instance, that we model the chronic diarrhoea incidence by multiplying the acute diarrhoea incidence by the probability of progressing from acute to chronic diarrhoea.

Possibly, epidemiological estimates can be derived through different approaches. In this case, cross-validation of the different estimates can be performed, which can strengthen their reliability.

Disease severity data

DALY estimates incorporate the severity of the health states through their duration and disability weight (DW)

[see Eq. (1)]. The former can be obtained through hospital registers, literature reviews or expert elicitation. The latter are most commonly derived from global burden of disease studies (Murray and Lopez 1996; Salomon et al. 2013) or the Dutch burden of disease study (Stouthard et al. 2000). Essink-Bot and Bonsel (2002) describe methods for DW derivation.

Step 4: data adjustment

Potential data biases should be critically appraised. Under-reporting and under-ascertainment are well-documented sources of bias (Lake et al. 2010). Misclassification bias due to imperfect diagnostic tests can be amended using various statistical techniques (Speybroeck et al. 2013). Finally, coherence in different epidemiological parameters for non-infectious diseases can be assessed using DisMod software (Barendregt et al. 2003).

Step 5: DALY calculation

Once all required data have been gathered, the actual DALY calculation can commence. DALYs for a disease, hazard or risk factor are obtained by summing the YLDs and YLLs calculated for each health state in the disease model. For a technical summary of DALY calculations, see Devleesschauwer et al. (2014). Despite wide availability of DALY calculation methods, standardized DALY calculation tools are currently not readily accessible. The World Health Organization provides a template for deterministic DALY calculations (available from <http://www.who.int/entity/healthinfo/bodreferencedalycalculationtemplate.xls>). Alternatively, the DALY calculator in R can be used for stochastic DALY calculations (Devleesschauwer et al. 2013). Until now, however, most users have made their own calculation model, either in a programming language such as R, or in Microsoft Excel with add-ins enabling Monte Carlo simulation. Although flexible, these methods do not guarantee methodological transparency. Further standardization of DALY calculation tools, including databases with default datasets, is therefore necessary to improve consistency in DALY calculations and to further DALY-based burden of disease studies.

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